Lecture 21

Inverse Laplace Transforms

So far we have seen how to find the Laplace transform of a variety of functions and derivatives. We can view the Laplace transform as an operator, that is, a function whose domain and range consists of functions.

$D : f \to f'$

The transform is defined as

$$\mathscr{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt \leftarrow$$

and we can write $\mathscr{L} : f(t) \to F(s)$ to say that $\mathscr{L}\{f(t)\} = F(s)$. We can then discuss the reverse operation or the *operator* we call the inverse Laplace transform

$$\mathscr{L}^{-1} : F(s) \to f(t)$$

Definition: Given a function $F(s)$, if there is a function $f(t)$ that is continuous on $[0, \infty)$ and satisfies

$$\mathscr{L}\{f(t)\} = F(s)$$

then $f(t)$ is the **inverse Laplace transform** of $F(s)$ and we write

$$f(t) = \underset{\substack{\uparrow \\ \text{operator}}}{\mathscr{L}^{-1}}\{F(s)\}$$

Example: Find the inverse Laplace transform of the following functions:

- $F(s) = \dfrac{6}{s^4}$

Q: What is $f(t)$ such that $\mathscr{L}\{f(t)\} = \frac{6}{s^4}$

Recall, $\mathscr{L}\{t^n\} = \dfrac{n!}{s^{n+1}}$

note $\dfrac{6}{s^4} = \dfrac{3!}{s^4} = \dfrac{3!}{s^{3+1}} \Rightarrow \mathscr{L}\{t^3\} = \dfrac{3!}{s^{3+1}} = \dfrac{6}{s^4} = F(s)$

So $\mathscr{L}^{-1}\left\{\dfrac{6}{s^4}\right\} = t^3$

- $F(s) = \dfrac{8}{s^2+64} = \dfrac{8}{s^2+8^2}$, $\mathcal{L}^{-1}\left\{\dfrac{8}{s^2+64}\right\} = \sin(8t)$

$b=8$

Recall,

$\mathcal{L}\{\sin(bt)\} = \dfrac{b}{s^2+b^2}$

$a=-2$ $\quad w = s+2$

- $F(s) = \dfrac{s+2}{s^2+4s+5} = \dfrac{s+2}{s^2+4s+4-4+5} = \dfrac{s+2}{(s+2)^2+1} = \dfrac{w}{w^2+1}$

$b=1$

Recall, $\mathcal{L}\{e^{at}f(t)\} = \mathcal{L}\{f(t)\}_{s\to s-a} = F(s-a)$

$\mathcal{L}\{\cos(bt)\} = \dfrac{s}{s^2+b^2}$

$\mathcal{L}^{-1}\left\{\dfrac{s+2}{s^2+4s+5}\right\} = e^{-2t}\cos(t)$

$\mathcal{L}\{e^{at}\cos(bt)\} = \dfrac{s-a}{(s-a)^2+b^2}$

Theorem (Linearity of the Inverse Transform): The inverse Laplace transform operator, $\mathcal{L}^{-1}$, is a linear operator. That is, suppose that $\mathcal{L}^{-1}\{F_1\}$ and $\mathcal{L}^{-1}\{F_2\}$ exist and are continuous on $[0,\infty)$ and c_1 and c_2 are constants, then

Recall, $\mathcal{L}\{e^{at}\} = \dfrac{1}{s-a}$

$$\mathcal{L}^{-1}\{c_1F_1 + c_2F_2\} = c_1\,\mathcal{L}^{-1}\{F_1\} + c_2\,\mathcal{L}^{-1}\{F_2\}$$

Example: Determine $\mathcal{L}^{-1}\left\{\dfrac{5}{s-6} - \dfrac{6s}{s^2+9} + \dfrac{3}{2s^2+8s+10}\right\}$

$= \mathcal{L}^{-1}\left\{\dfrac{5}{s-6}\right\} - \mathcal{L}^{-1}\left\{\dfrac{6s}{s^2+9}\right\} + \mathcal{L}^{-1}\left\{\dfrac{3}{2s^2+8s+10}\right\}$

$= 5\cdot\mathcal{L}^{-1}\left\{\dfrac{1}{s-6}\right\} - 6\mathcal{L}^{-1}\left\{\dfrac{s}{s^2+9}\right\} + 3\mathcal{L}^{-1}\left\{\dfrac{1}{2(s^2+4s+5)}\right\}$

$= 5\cdot e^{6t} - 6\cos(3t) + \frac{3}{2}\mathcal{L}^{-1}\left\{\dfrac{1}{(s+2)^2+1}\right\}$

$= 5e^{6t} - 6\cos(3t) + \frac{3}{2}e^{-2t}\sin(t)$

Recall $\mathcal{L}\{e^{at}\sin(bt)\} = \dfrac{b}{(s-a)^2+b^2}$

$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$

$e^{at}f(t) = \mathcal{L}^{-1}\{F(s-a)\} = e^{at}\mathcal{L}^{-1}\{F(s)\}$

Example: Determine $\mathcal{L}^{-1}\left\{\dfrac{5}{(s-2)^5}\right\} = e^{2t}\mathcal{L}^{-1}\left\{\dfrac{5}{s^5}\right\}$

$\mathcal{L}\{t^n\} = \dfrac{n!}{s^{n+1}}$

$\mathcal{L}\{t^4\} = \dfrac{4!}{s^5} = \dfrac{24}{s^5}$

$= 5e^{2t}\mathcal{L}^{-1}\left\{\dfrac{24}{s^5}\cdot\dfrac{1}{24}\right\}$

$= \boxed{\dfrac{5}{24}e^{2t}t^4}$

Consider the situation where we want to find the inverse transform of the following two functions

$$F_1(s) = \frac{7s^2+10s-1}{s^3+3s^2-s-3} \text{ and } F_2(s) = \frac{2}{s-1} + \frac{1}{s+1} + \frac{4}{s+3}$$

it would be much easier to find the inverse transform of F_2. It can be verified that $\boxed{F_1 = F_2}$. One can convert F_1 into F_2 through the use of a **partial fraction decomposition**. We will see that partial fraction decomposition is important in solving equations using Laplace transforms and inverse Laplace transforms.

We will review partial fraction decomposition before moving on to solving equations.

$$\begin{array}{r} x \\ x^2+1 \overline{)\, x^3+1} \\ -(x^3+x) \\ 1-x \end{array}$$

$$\frac{x^3+1}{x^2+1} = x + \frac{1-x}{x^2+1}$$

Method of Partial Fractions

Given the fraction $\dfrac{P(s)}{Q(s)}$, where $P(s)$ and $Q(s)$ are polynomials with $\deg(P) < \deg(Q)$. $Q(s)$ can always be factored into a product of linear or irreducible quadratic factors. We can break down the factorization of $Q(s)$ into 3 separate cases.

Case 1: Nonrepeated Linear Factors:

Suppose that $Q(s) = (s-r_1)(s-r_2)\cdots(s-r_n)$ where $r_1, r_2, \ldots, r_n$ are distinct real numbers. Then we have the decomposition

$$\frac{P(s)}{Q(s)} = \frac{A_1}{s-r_1} + \frac{A_2}{s-r_2} + \cdots + \frac{A_n}{s-r_n}$$

where $A_1, A_2, \ldots, A_n$ are real numbers.

Example: Consider $F(s) = \dfrac{1}{(s-3)(s+1)} = \left(\dfrac{A_1}{s-3} + \dfrac{A_2}{s+1}\right)$

$\Rightarrow \quad 1 = A_1(s+1) + A_2(s-3)$

Let $s=3$: $1 = A_1(4) + A_2(0) \Rightarrow A_1 = \frac{1}{4}$

$s=-1$: $1 = A_1(0) + A_2(-4) \Rightarrow A_2 = -\frac{1}{4}$

$$\frac{1}{(s-3)(s+1)} = \frac{1/4}{s-3} + \frac{-1/4}{s+1}$$

Case 2: Repeated Linear Factors: Ex: $\dfrac{1}{(s-3)^2(s+1)}$

If $Q(s)$ contains the factor $(s-r)^m$ where m is the multiplicity of the root r in $Q(s)$, then the portion of the decomposition corresponding to $(s-r)^m$ is

$$\frac{A_1}{s-r} + \frac{A_2}{(s-r)^2} + \cdots + \frac{A_m}{(s-r)^m}$$

where $A_1, A_2, \ldots, A_m$ are real numbers.

Example: Consider $F(s) = \dfrac{1}{(s-3)^2(s+1)} = \left(\dfrac{A_1}{s-3} + \dfrac{A_2}{(s-3)^2}\right) + \dfrac{A_3}{s+1}$

$$\Rightarrow \quad 1 = A_1(s-3)(s+1) + A_2(s+1) + A_3(s-3)^2$$

Let $s=3$: $1 = A_1(0) + A_2(4) + A_3(0) \Rightarrow \boxed{A_2 = \frac{1}{4}}$

$s=-1$: $1 = A_1(0) + A_2(0) + A_3(-4)^2$

$$\Rightarrow \quad 1 = 16A_3 \Rightarrow \boxed{A_3 = \frac{1}{16}}$$

choose anything

$s=0$: $1 = A_1(-3)(1) + \frac{1}{4}(1) + \frac{1}{16}(-3)^2$

$$\Rightarrow \quad 1 = -3A_1 + \frac{1}{4} + \frac{9}{16}$$

$$\Rightarrow \quad 1 - \frac{1}{4} - \frac{9}{16} = -3A_1 \Rightarrow \frac{3}{16} = -3A_1$$

$$\Rightarrow \quad \boxed{A_1 = -\frac{1}{16}}$$

So $$\frac{1}{(s-3)^2(s+1)} = \frac{-1/16}{s-3} + \frac{1/4}{(s-3)^2} + \frac{1/16}{s+1}$$

Case 3: Irreducible Quadratic Factors:

$s^2+\beta^2,\ \alpha=0$

$s^2-2\alpha s+\beta^2+\alpha^2$

If $Q(s)$ contains the factor $((s-\alpha)^2+\beta^2)^m$, then the portion of the decomposition corresponding to this factor is

$$\frac{A_1(s-\alpha)+B_1\beta}{(s-\alpha)^2+\beta^2}+\frac{A_2(s-\alpha)+B_2\beta}{[(s-\alpha)^2+\beta^2]^2}+\cdots+\frac{A_m(s-\alpha)+B_m}{[(s-\alpha)^2+\beta^2]^m}$$

Example: Consider $F(s)=\dfrac{2s^2+10s}{(s+1)(s^2-2s+5)}$

$s^2-2s+1-1+5$

$$\frac{A_1(s-\alpha)+B_1\beta}{(s-\alpha)^2+\beta^2}=A_1\frac{s-\alpha}{(s-\alpha)^2+\beta^2}+B_1\frac{\beta}{(s-\alpha)^2+\beta^2}$$

$$=\frac{2s^2+10s}{(s+1)((s-1)^2+4)}$$

$\alpha=1$, $\beta=2$

$$=\frac{A_1}{s+1}+\frac{A_2(s-1)+B_2\cdot 2}{(s-1)^2+4}$$

$\Rightarrow 2s^2+10s=A_1((s-1)^2+4)+(A_2(s-1)+B_2\cdot 2)(s+1)$

Let $s=-1$: $2(-1)^2+10(-1)=A_1((-2)^2+4)+0$

$\Rightarrow -8=8A_1 \Rightarrow \boxed{A_1=-1}$

Let $s=1$: $2+10=-1(0+4)+(A_2(0)+2B_2)(2)$

$\Rightarrow 12=-4+4B_2 \Rightarrow 4B_2=16 \Rightarrow \boxed{B_2=4}$

Let $s=0$: $0=-1(5)+(A_2(-1)+8)(1)$

$\Rightarrow 0=-5-A_2+8 \Rightarrow -A_2=-3$

$\Rightarrow \boxed{A_2=3}$

So $\dfrac{2s^2+10s}{(s+1)(s^2-2s+5)}=\dfrac{-1}{s+1}+\dfrac{3(s-1)+4\cdot 2}{(s-1)^2+4}$

Example: Determine both $\boxed{\mathscr{L}^{-1}\{\ln(s-a)\}}$ and $\mathscr{L}^{-1}\{\arctan(s)\}$.

$$\boxed{\mathscr{L}^{-1}\{\ln(s-a)\} = -\frac{e^{at}}{t}}$$

$$\frac{d}{ds}\ln(s-a) = \frac{1}{s-a}$$

$$\mathscr{L}\{tf(t)\} = -\frac{d}{ds}F(s)$$

$$\frac{d}{ds}\ln(s-a) \quad \mathscr{L}\{-tf(t)\}$$

$$\frac{d}{ds}\ln(s-a) = \frac{1}{s-a}$$

$$-tf(t) = \mathscr{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$$

$$\Rightarrow f(t) = -\frac{e^{at}}{t}$$

$$\frac{1}{s^2+1} = \frac{d}{ds}\arctan(s) = \mathscr{L}\{-tf(t)\}$$

$$-tf(t) = \mathscr{L}^{-1}\left\{\frac{1}{s^2+1}\right\}$$

$$= \sin(t)$$

$$\Rightarrow \boxed{\mathscr{L}^{-1}\{\arctan(s)\} = -\frac{\sin(t)}{t}}$$

With all of these tools we can start to solve linear initial value problems. To do so, we can use the following steps:

1. Apply the Laplace transform to both sides of the equation.
2. Convert into an equation for $Y(s) = \mathscr{L}\{y(t)\}$.
3. Solve for $Y(s)$
4. Use the inverse transform to find $y(t)$.

Recall

$$\mathscr{L}\{f'\} = sF(s) - f(0)$$

Example: Solve $y' - 3y = e^{2t}$, $y(0) = 1$

$$\mathscr{L}\{y' - 3y\} = \mathscr{L}\{e^{2t}\}$$

$$\Rightarrow sY(s) - y(0) - 3Y(s) = \frac{1}{s-2}$$

$$\Rightarrow (s-3)Y(s) - 1 = \frac{1}{s-2} \Rightarrow (s-3)Y(s) = \frac{1}{s-2} + 1$$

$$\Rightarrow Y(s) = \frac{1}{(s-2)(s-3)} + \frac{1}{s-3}$$

$$= \underbrace{\left(\frac{-1}{s-2} + \frac{1}{s-3}\right)}_{\downarrow\, e^{3t}} + \underbrace{\frac{1}{s-3}}_{\downarrow\, e^{3t}}$$

PFD: $\frac{1}{(s-2)(s-3)} = \frac{A_1}{s-2} + \frac{A_2}{s-3}$

$$\Rightarrow A_1 = -1,\ A_2 = 1$$

$$\Rightarrow y(t) = \mathscr{L}^{-1}\{\}$$

$$= \boxed{-1e^{2t} + 2e^{3t}}$$

Example: Solve $y''' - y'' + y' - y = 0$, $y(0) = 1$, $y'(0) = 1$, $y''(0) = 3$

$$\mathcal{L}\{y'''\} = s^3 Y(s) - s^2 y(0) - s y'(0) - y''(0)$$

$$\mathcal{L}\{y''\} = s^2 Y(s) - s y(0) - y'(0)$$

$$\mathcal{L}\{y'\} = s Y(s) - y(0)$$

$$\mathcal{L}\{y''' - y'' + y' - y = 0\} \qquad \mathcal{L}\{0\} = 0$$

$$\Rightarrow (s^3 Y(s) - s^2 - \cancel{s} - 3) - (s^2 Y(s) - \cancel{s} - 1) + (sY(s) - 1) - Y(s) = 0$$

$$\Rightarrow Y(s)\underbrace{(s^3 - s^2 + s - 1)}_{\text{char eqn}} - s^2 - 3 = 0$$

$$\Rightarrow Y(s)(s^3 - s^2 + s - 1) = s^2 + 3$$

$$\Rightarrow Y(s) = \frac{s^2+3}{s^3 - s^2 + s - 1} = \frac{s^2+3}{s^2(s-1) + 1(s-1)} = \frac{s^2+3}{(s^2+1)(s-1)}$$

$$\Rightarrow Y(s) = \frac{s^2+3}{(s^2+1)(s-1)} = \frac{A_1 s + B_1}{s^2+1} + \frac{A_2}{s-1}$$

$$\Rightarrow s^2 + 3 = (A_1 s + B_1)(s-1) + A_2(s^2+1)$$

$A_2 = 2$
$B_1 = -1$
$A_1 = -1$

$s = -1$
$4 = (-A_1 - 1)(-2) + A_2(2)$
$4 = 2A_1 + 2 + 4$
$A_1 = -1$

$$\Rightarrow Y(s) = \frac{-s-1}{s^2+1} + \frac{2}{s-1}$$

$$= -\frac{s}{s^2+1} - \frac{1}{s^2+1} + \frac{2}{s-1}$$

$$\Rightarrow \boxed{y(t) = -\cos(t) - \sin(t) + 2e^t}$$

$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0)$

not centered at 0

Example: Solve $w'' - 2w' + w = 6t - 2$, $w(-1) = 3$, $w'(-1) = 7$

Use $y(t) = w(t-1) \Rightarrow y(0) = w(0-1) = w(-1) = 3$

$t=0$

$y'(0) = w'(-1) = 7$

$\Rightarrow w''(t-1) - 2w'(t-1) + w(t-1) = 6(t-1) - 2$

$\Rightarrow$ new IVP $y'' - 2y' + y = 6t - 8$, $y(0) = 3$, $y'(0) = 7$

$$s^2Y(s) - 3s - 7 - 2sY(s) + 6 + Y(s) = \frac{6}{s^2} - \frac{8}{s} = \frac{6s-8}{s^2}$$

$$Y(s)(s^2 - 2s + 1) - 3s - 1 = \frac{6s-8}{s^2}$$

combine

$$\Rightarrow Y(s) = \frac{\frac{6s-8}{s^2} + 3s + 1}{s^2 - 2s + 1} = \frac{\frac{3s^3 + s^2 + 6s - 8}{s^2}}{\underbrace{s^2 - 2s + 1}_{(s-1)^2}}$$

$$\Rightarrow Y(s) = \frac{3s^3 + s^2 + 6s - 8}{s^2(s-1)^2} \overset{\text{PFD}}{=} \frac{A_1}{s} + \frac{A_2}{s^2} + \frac{A_3}{(s-1)} + \frac{A_4}{(s-1)^2}$$

$\Rightarrow A_1 = 4,\ A_2 = 6,\ A_3 = -1,\ A_4 = 2$

$$Y(s) = \frac{4}{s} + \frac{6}{s^2} + \frac{-1}{s-1} + \frac{2}{(s-1)^2} \leftarrow e^t\mathcal{L}^{-1}\left\{\frac{2}{s^2}\right\}$$

$$\Rightarrow \boxed{y(t) = 4 + 6t - 1e^t + 2e^t \cdot t}$$

Example: Solve $y'' + ty' - y = 0$, $y(0) = 0$, $y'(0) = 3$

Recall $\mathcal{L}\{tf(t)\} = -\frac{d}{ds}F(s)$,

$\mathcal{L}\{ty\} = -Y'(s)$

$\mathcal{L}\{ty'\} = -\frac{d}{ds}\mathcal{L}\{y'\}$

$= -\frac{d}{ds}(sY(s) - y(0))$

$= -Y(s) - sY'(s)$

$\mathcal{L}\{ty''\} = -\frac{d}{ds}(s^2Y(s) - sy(0) - y'(0))$

$= -2sY(s) - s^2Y'(s) + y(0)$

$(s^2Y(s) - s(0) - 3) + (-Y(s) - sY'(s)) - Y(s) = 0$

$-sY'(s) + Y(s)(s^2 - 2) - 3 = 0$

$\Rightarrow Y'(s) + \frac{(s^2-2)}{-s}Y(s) = \frac{3}{-s}$

$\Rightarrow Y'(s) + \left(\frac{2}{s} - s\right)Y(s) = -\frac{3}{s}$

integrating factor: $P(s) = \frac{2}{s} - s$, $Q(s) = -\frac{3}{s}$

$\mu(s) = e^{\int P(s)ds} = e^{\int \frac{2}{s} - s\,ds} = e^{2\ln(s) - \frac{1}{2}s^2} = s^2e^{-1/2 s^2}$

$Y(s) = \frac{1}{\mu(s)}\int \mu(s)Q(s)\,ds = \frac{1}{s^2e^{-s^2/2}}\int s^2e^{-s^2/2} \cdot -\frac{3}{s}\,ds$

$= \frac{1}{s^2e^{-s^2/2}}\int -3se^{-s^2/2}\,ds$

$= \frac{1}{s^2e^{-s^2/2}}\left(\int 3e^u\,du = \frac{1}{s^2e^{-s^2/2}}(3e^{-s^2/2} + C)\right)$ $\quad u = \frac{-s^2}{2}$, $du = -s\,ds$

$\Rightarrow Y(s) = \frac{3}{s^2} + \frac{C}{s^2} \cdot e^{s^2/2}$

Recall: For $\mathcal{L}\{f(t)\} = F(s)$, f piecewise cont. and exp. order α

$\lim_{s\to\infty} F(s) = 0 \Rightarrow \lim_{s\to\infty} Y(s) = 0 \Rightarrow \lim_{s\to\infty}\left[\underbrace{\frac{3}{s^2}}_{\to 0} + \underbrace{\frac{C}{s^2e^{s^2/2}}}_{\to\infty}\right] = 0$

$\Rightarrow C = 0$

So $Y(s) = \frac{3}{s^2} \Rightarrow \boxed{y(t) = 3t}$

Example: Solve $ty'' - ty' + y = 2$, $y(0) = 2$, $y'(0) = -1$